RTL WITH VERILOG LAB 3

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Q1) Modified Traffic Light Controller

Design Code:

/home/cl4/RTL_WITH_VERLILOG3/modified_traffic_light_controller.v - Default

```
module modified_traffic_light_controller (
input clk,
input reset,
input button,
input ped_button,
output reg red,
output reg yellow,
output reg green
);
localparam RED_STATE = 2'b00,
GREEN_STATE = 2'b01,
YELLOW_STATE = 2'b10,
PED_STATE = 2'b11;
reg [1:0] current_state;
reg [1:0] next_state;
reg ped_request;
always @(posedge clk) begin
if (reset) begin
current_state <= RED_STATE;
ped_request <= 1'b0;
end
else begin
if (ped_button)
ped_request <= 1'b1;
if (button) begin
current_state <= next_state;
if (current_state == PED_STATE)
ped_request <= 1'b0;
end
end
end
always @(*) begin
case (current_state)
RED_STATE: begin
if (ped_request)
next_state = PED_STATE;
else
next_state = GREEN_STATE;
```

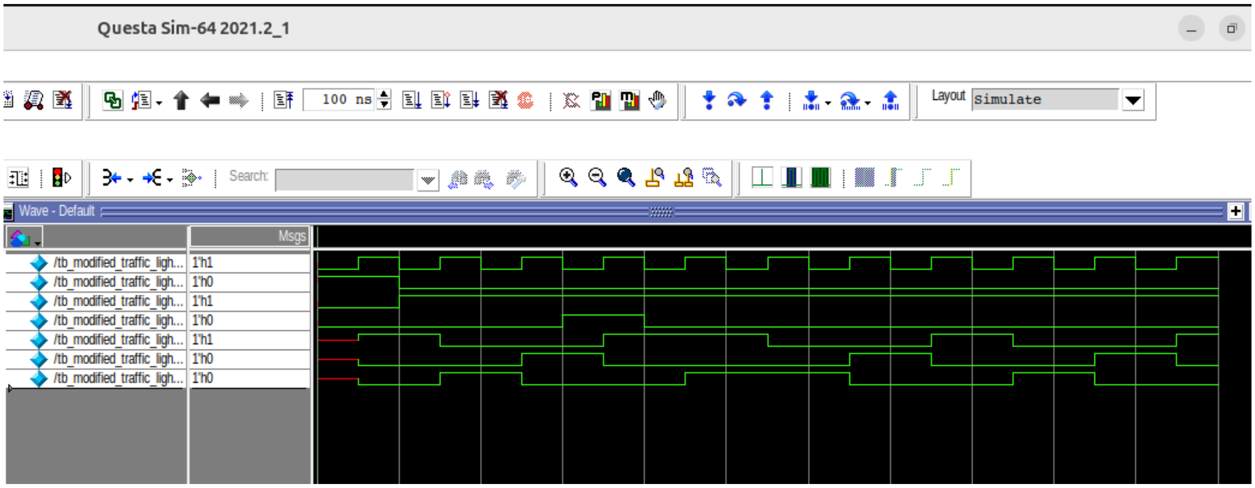
```
end
GREEN_STATE:
next_state = YELLOW_STATE;
YELLOW_STATE:
next_state = RED_STATE;
PED_STATE:
next_state = GREEN_STATE;
default:
next_state = RED_STATE;
endcase
end
always @(*) begin
red = 1'b0;
yellow = 1'b0;
green = 1'b0;
case (current_state)
RED_STATE:
red = 1'b1;
GREEN_STATE:
green = 1'b1;
YELLOW_STATE:
yellow = 1'b1;
PED_STATE: begin
red = 1'b1;
green = 1'b1;
end
endcase
end
endmodule
```

Testbench Code:

/home/cl4/RTL_WITH_VERLILOG3/tb_modified_traffic_light_controller.v - Default

```verilog
module tb_modified_traffic_light_controller;
reg clk;
reg reset;
reg button;
reg ped_button;
wire red;
wire yellow;
wire green;
modified_traffic_light_controller uut (
.clk(clk),
.reset(reset),
.button(button),
.ped_button(ped_button),
.red(red),
.yellow(yellow),
.green(green)
);
always #5 clk = ~clk;
initial begin
clk = 0;
reset = 1;
button = 0;
ped_button = 0;
$monitor("Time=%0t | Reset=%b Button=%b Ped_Button=%b | Red=%b Yellow=%b Green=%b
$time, reset, button, ped_button, red, yellow, green);
#10;
reset = 0;
button = 1; #20;
ped_button = 1; #10;
ped_button = 0; #30;
button = 1; #40;
$finish;
end
endmodule
```

QuestaSim Waveform:



FPGA Commands:

```
cl4@CL4-29: ~/f4pga-examples/xc7
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C mod_traffic clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
/home/cl4/f4pga-examples/xc7/mod_traffic/../../common/common.mk:39: *** Unsupported board type.  Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C mod_traffic
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_traffic'
mkdir -p /home/cl4/f4pga-examples/xc7/mod_traffic/build/arty_100
cd /home/cl4/f4pga-examples/xc7/mod_traffic/build/arty_100 && symbiflow_synth -t modified_traffic_light_control
t_controller.v -d artix7 -p xc7a100tcsg324-1 -x /home/cl4/f4pga-examples/xc7/mod_traffic/arty.xdc
[F4PGA] Running (deprecated) synth
-t
modified_traffic_light_controller
-v
/home/cl4/f4pga-examples/xc7/mod_traffic/modified_traffic_light_controller.v
-d
artix7
```

Q2) Vending Machine
Design Code:

/home/cl4/RTL_WITH_VERLILOG3/vending_machine.v - Default

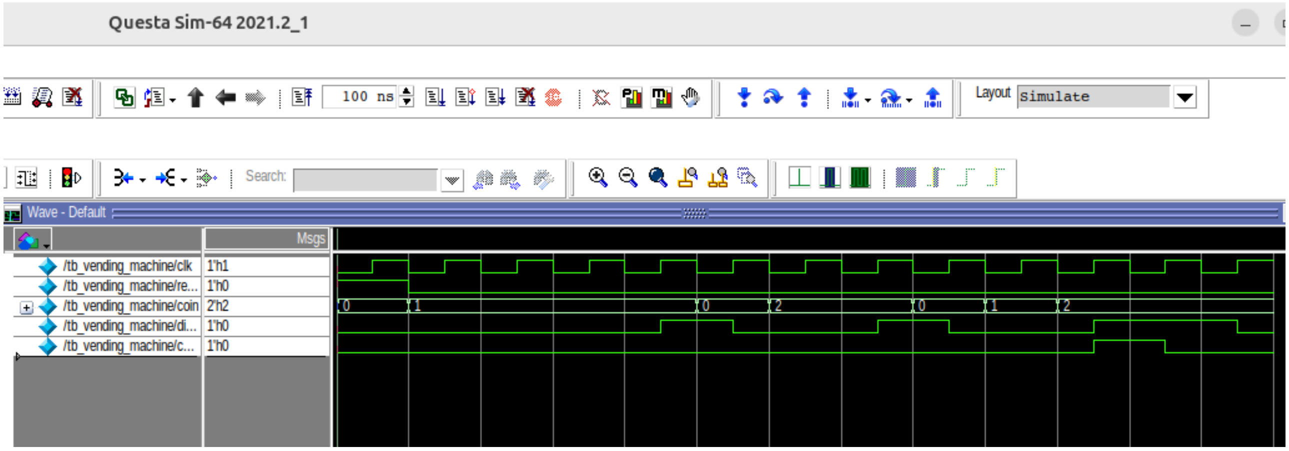
```
module vending_machine (
input clk,
input reset,
input [1:0] coin,
output reg dispense,
output reg change
);
localparam S0 = 3'b000,
S5 = 3'b001,
S10 = 3'b010,
S15 = 3'b011,
S20 = 3'b100;
reg [2:0] current_state;
reg [2:0] next_state;

always @(posedge clk or posedge reset) begin
if (reset)
current_state <= S0;
else
current_state <= next_state;
end

always @(*) begin
case (current_state)
S0: begin
if (coin == 2'b01)
next_state = S5;
else if (coin == 2'b10)
next_state = S10;
else
next_state = S0;
end
S5: begin
if (coin == 2'b01)
next_state = S10;
else if (coin == 2'b10)
next_state = S15;
else
```

```
next_state = S5;
end
S10: begin
if (coin == 2'b01)
next_state = S15;
else if (coin == 2'b10)
next_state = S20;
else
next_state = S10;
end
S15: begin
if (coin == 2'b01)
next_state = S20;
else if (coin == 2'b10)
next_state = S20;
else
next_state = S15;
end
S20: next_state = S0;
default: next_state = S0;
endcase
end

always @(*) begin
dispense = 1'b0;
change = 1'b0;
case (current_state)
S15: begin
if (coin == 2'b10) begin
dispense = 1'b1;
change = 1'b1;
end
end
S20: begin
dispense = 1'b1;
end
endcase
end
endmodule
```

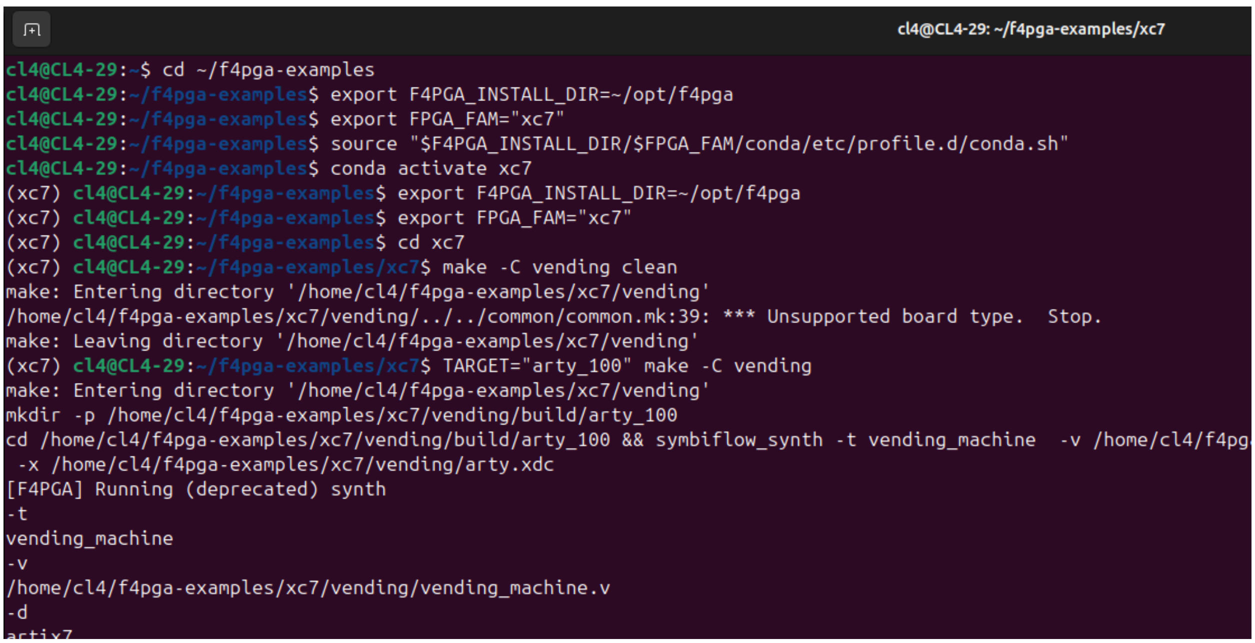
Testbench Code:

```
module tb_vending_machine;
reg clk;
reg reset;
reg [1:0] coin;
wire dispense;
wire change;

vending_machine uut (
.clk(clk),
.reset(reset),
.coin(coin),
.dispense(dispense),
.change(change)
);

always #5 clk = ~clk;
initial begin
clk = 0;
reset = 1;
coin = 2'b00;
$monitor("Time=%0t | Reset=%b Coin=%b | State=%d Dispense=%b Change=%b",
$time, reset, coin, uut.current_state, dispense, change);
#10;
reset = 0;
coin = 2'b01; #10;
coin = 2'b01; #10;
coin = 2'b01; #10;
coin = 2'b01; #10;
coin = 2'b00; #10;
coin = 2'b10; #10;
coin = 2'b10; #10;
coin = 2'b00; #10;
coin = 2'b01; #10;
coin = 2'b10; #10;
coin = 2'b10; #20;
$finish;
end
endmodule
```

QuestaSim Waveform:



FPGA Commands:



```
cl4@CL4-29: ~/f4pga-examples/xc7
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C vending clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/vending'
/home/cl4/f4pga-examples/xc7/vending/../../common/common.mk:39: *** Unsupported board type.  Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/vending'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C vending
make: Entering directory '/home/cl4/f4pga-examples/xc7/vending'
mkdir -p /home/cl4/f4pga-examples/xc7/vending/build/arty_100
cd /home/cl4/f4pga-examples/xc7/vending/build/arty_100 && symbiflow_synth -t vending_machine  -v /home/cl4/f4pg
 -x /home/cl4/f4pga-examples/xc7/vending/arty.xdc
[F4PGA] Running (deprecated) synth
-t
vending_machine
-v
/home/cl4/f4pga-examples/xc7/vending/vending_machine.v
-d
artix7
```

Q3) Modified Synchronous FIFO

Design Code:

/home/cl4/RTL_WITH_VERLILOG3/modified_synchronous_fifo.v - Default

Ln#

```verilog
module modified_synchronous_fifo (
input clk,
input reset,
input write_enable,
input read_enable,
input [1:0] data_in,
output reg [1:0] data_out,
output full,
output empty,
output almost_full,
output almost_empty
);
parameter DEPTH = 4;
reg [1:0] memory [0:DEPTH-1];
reg [1:0] write_pointer;
reg [1:0] read_pointer;
reg [2:0] count;

always @(posedge clk) begin
if (reset) begin
write_pointer <= 2'd0;
read_pointer <= 2'd0;
count <= 3'd0;
data_out <= 2'd0;
end
else begin
case ({write_enable && !full, read_enable && !empty})
2'b10: begin
memory[write_pointer] <= data_in;
write_pointer <= write_pointer + 1'b1;
count <= count + 1'b1;
end
2'b01: begin
data_out <= memory[read_pointer];
read_pointer <= read_pointer + 1'b1;
count <= count - 1'b1;
end
```

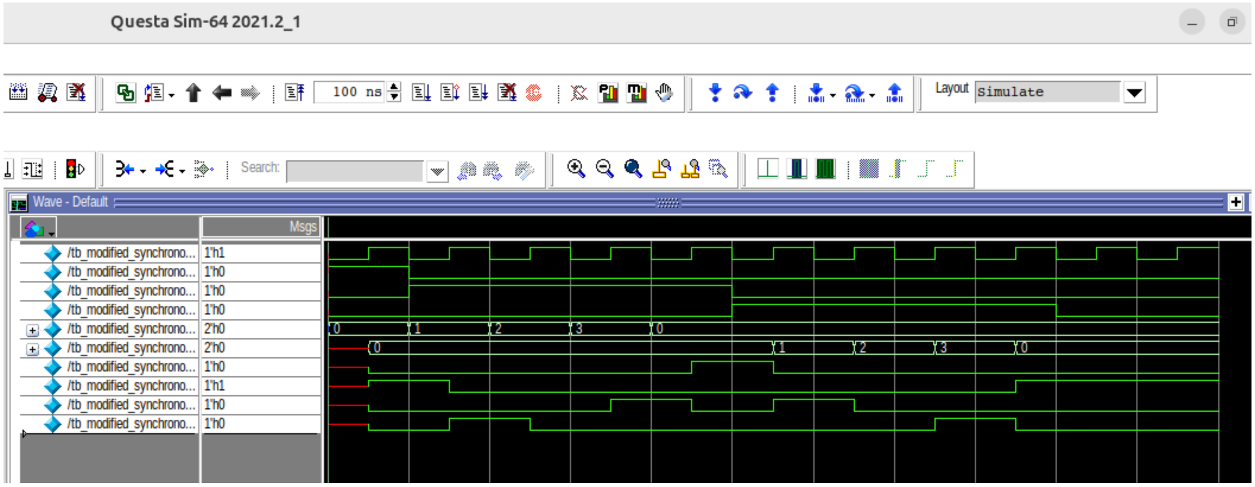
```
2'b11: begin
memory[write_pointer] <= data_in;
data_out <= memory[read_pointer];
write_pointer <= write_pointer + 1'b1;
read_pointer <= read_pointer + 1'b1;
end
default: ;
endcase
end
end

assign full = (count == DEPTH);
assign empty = (count == 0);
assign almost_full = (count == DEPTH - 1);
assign almost_empty = (count == 1);
endmodule
```

Testbench Code:

/home/cl4/RTL_WITH_VERLILOG3/tb_modified_synchronous_fifo.v - Default

```
module tb_modified_synchronous_fifo;
reg clk;
reg reset;
reg write_enable;
reg read_enable;
reg [1:0] data_in;
wire [1:0] data_out;
wire full;
wire empty;
wire almost_full;
wire almost_empty;

modified_synchronous_fifo uut (
.clk(clk),
.reset(reset),
.write_enable(write_enable),
.read_enable(read_enable),
.data_in(data_in),
.data_out(data_out),
.full(full),
.empty(empty),
.almost_full(almost_full),
.almost_empty(almost_empty)
);

always #5 clk = ~clk;
initial begin
clk = 0;
reset = 1;
write_enable = 0;
read_enable = 0;
data_in = 2'd0;
$monitor("Time=%0t | WE=%b RE=%b DataIn=%d DataOut=%d | Full=%b AF=%b AE=%b Empty=%b",
$time, write_enable, read_enable, data_in, data_out, full, almost_full, almost_empty, empty);
#10;
reset = 0;
write_enable = 1;
data_in = 2'd1; #10;
data_in = 2'd2; #10;
data_in = 2'd3; #10;
data_in = 2'd0; #10;
write_enable = 0;
read_enable = 1; #10;
read_enable = 1; #10;
read_enable = 1; #10;
read_enable = 1; #10;
read_enable = 0; #20;
$finish;
end
endmodule
```

QuestaSim Waveform:



FPGA Commands:

cl4@CL4-29: ~/f4pga-examples/xc7

```
cl4@CL4-29:~$ cd ~/f4pga-examples
cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
cl4@CL4-29:~/f4pga-examples$ export FPGA_FAM="xc7"
cl4@CL4-29:~/f4pga-examples$ source "$F4PGA_INSTALL_DIR/$FPGA_FAM/conda/etc/profile.d/conda.sh"
cl4@CL4-29:~/f4pga-examples$ conda activate xc7
(xc7) cl4@CL4-29:~/f4pga-examples$ export F4PGA_INSTALL_DIR=~/opt/f4pga
(xc7) cl4@CL4-29:~/f4pga-examples$ cd ~/f4pga-examples
(xc7) cl4@CL4-29:~/f4pga-examples$ cd xc7
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ make -C mod_fifo clean
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
/home/cl4/f4pga-examples/xc7/mod_fifo/../../common/common.mk:39: *** Unsupported board type.  Stop.
make: Leaving directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
(xc7) cl4@CL4-29:~/f4pga-examples/xc7$ TARGET="arty_100" make -C mod_fifo
make: Entering directory '/home/cl4/f4pga-examples/xc7/mod_fifo'
mkdir -p /home/cl4/f4pga-examples/xc7/mod_fifo/build/arty_100
cd /home/cl4/f4pga-examples/xc7/mod_fifo/build/arty_100 && symbiflow_synth -t modified_synchronous_fifo  -v /ho
x7 -p xc7a100tcsg324-1 -x /home/cl4/f4pga-examples/xc7/mod_fifo/arty.xdc
[F4PGA] Running (deprecated) synth
-t
modified_synchronous_fifo
-v
/home/cl4/f4pga-examples/xc7/mod_fifo/modified_synchronous_fifo.v
-d
artix7
```